A **survey** on technologies for automatic **forest fire** monitoring, detection, and fighting using unmanned aerial vehicles and remote sensing techniques

C Yuan, Y Zhang, Z Liu - Canadian journal of forest research, 2015 - cdnsciencepub.com

… and tracking of the **propagation** of large forest fires. A numerical **propagation model** for **forest fire** monitoring and detection was verified in **simulation** using a dynamic UAV **model** with six …

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A household-level approach to staging **wildfire** evacuation warnings using trigger modeling

D Li, TJ Cova, PE Dennison - Computers, Environment and Urban Systems, 2015 - Elsevier

… GIS) and fire-**spread** modeling to calculate evacuation trigger … scenarios including dynamic **forecast** weather conditions, … This work explores the potential uses of **wildfire** evacuation …

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Machine learning based **forest fire susceptibility** assessment of Manavgat district (Antalya), Turkey

HA Akıncı, H Akıncı - Earth Science Informatics, 2023 - Springer

… Meanwhile, literature **review** showed that most FFSM studies … is to produce **forest fire susceptibility maps** for the forests in … Slope affects the **spread** of fire. On steeper slopes, the rate of …

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The fire and smoke **model** evaluation experiment—a plan for integrated, large fire–atmosphere field campaigns

S Prichard, NS Larkin, R Ottmar, NHF French, K Baker… - Atmosphere, 2019 - mdpi.com

… While the rate of **spread** can be **mapped** from high quality, … , fuel consumption, **wildland fire spread**, and smoke **prediction** for … and is an active participant in project planning and **review**. …

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Regional **forest-fire susceptibility** analysis in central Portugal using a probabilistic ratings procedure and artificial neural network weights assignment

LA Dimuccio, R Ferreira, L Cunha… - … of **Wildland Fire**, 2011 - connectsci.au

… to suffer a **forest fire** or to support its **spread** and is determined … , because our goal was to **map susceptibility** in the long term … We also acknowledge and appreciate the **review** of the first …

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Ubiquitous GIS-based **forest fire susceptibility mapping** using artificial intelligence methods

SV Razavi-Termeh, A Sadeghi-Niaraki, SM Choi - Remote Sensing, 2020 - mdpi.com

… The **model** uses an error **propagation** algorithm and a hybrid to train the network, reduce the … **Wildfire** field **survey** using mobile GIS technology in Nan province. In Proceedings of the …

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Forest fuel **mapping** and evaluation of LANDFIRE fuel **maps** in Boulder County, Colorado, USA

K Krasnow, T Schoennagel, TT Veblen - Forest Ecology and **Management**, 2009 - Elsevier

… A key challenge in modern **wildfire** mitigation and forest **management** is accurate **mapping** of … predictors will more accurately simulate fire **spread** rates and reflect the inherent variability …

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Sensitivity analysis of fire **behavior** modeling with LIDAR-derived surface fuel **maps**

M Mutlu, SC Popescu, K Zhao - Forest Ecology and **Management**, 2008 - Elsevier

… **model** fire **behavior** using FARSITE (Fire Area Simulator) and investigate differences in modeling outputs using fuel **model maps**, … + NVC **model** has the potential to minimize **wildfire** cost …

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Controls on the spatial pattern of **wildfire** ignitions in Southern California

N Faivre, Y Jin, ML Goulden… - … Journal of **Wildland Fire**, 2014 - CSIRO Publishing

… **spread** and recurrence, but information on how humans influence ignition patterns is still lacking at a scale compatible with integrated fire **management**. … The predictive **maps** of ignition …

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Evaluating **wildland fire** danger and prioritizing vegetation and fuels treatments

PF Hessburg, KM Reynolds, RE Keane… - … and **Management**, 2007 - Elsevier

We present a decision support application that evaluates danger of severe **wildland fire** and … **model**, we evaluate fire danger as a function of three primary topics: fire hazard, fire **behavior**…

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Characterizing and **mapping forest fire** fuels using ASTER imagery and gradient modeling

MJ Falkowski, PE Gessler, P Morgan, AT Hudak… - … and **management**, 2005 - Elsevier

… , for **mapping** fuel layers for fire **behavior** modeling with … the efficacy of ASTER for **mapping** and characterizing crown … (National **Forest Fire** Laboratory (NFFL) 1–13) were **mapped** …

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A neural network **model** for **wildfire** scale **prediction** using meteorological factors

H Liang, M Zhang, H Wang - IEEE Access, 2019 - ieeexplore.ieee.org

… The associate editor coordinating the **review** of this … two processes: forward **propagation** of information and backprop… additivity to predict fire frequency from **maps** of fire weather and land …

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Wildland fire detection and monitoring using a drone-collected rgb/ir image dataset

X Chen, B Hopkins, H Wang, L O'Neill, F Afghah… - IEEE …, 2022 - ieeexplore.ieee.org

… **wildfire**'s extent, **behavior**, and conditions in the fire's near environment, particularly during its early growth. Rapid **mapping** … of drones in remote **wildfire management** operations [12], [13]…

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Landscape–**wildfire** interactions in southern Europe: implications for landscape **management**

F Moreira, O Viedma, M Arianoutsou, T Curt… - … **management**, 2011 - Elsevier

… In this paper we **review** the available scientific knowledge on the … Solid lines indicate the main drivers of fire spatial **spread** and … **maps** have been the most common sources of data used. …

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Integrated **wildfire** risk assessment: Framework development and application on the Lewis and Clark National Forest in Montana, USA

MP Thompson, J Scott, D Helmbrecht… - … and **management**, 2013 - academic.oup.com

… A prerequisite step for modeling **wildfire** growth and **behavior** is **mapping** the vegetative and fuel … Below we briefly **review** each step in the context of the Little Belts Assessment. …

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Determination of **forest fire** risk with fuzzy analytic hierarchy process and its **mapping** with the application of GIS: The case of Turkey/Çakırlar

C Güngöroğlu - Human and Ecological Risk Assessment: An …, 2017 - Taylor & Francis

… Furthermore, high-accuracy estimation regarding **forest fire behavior** is made difficult by the … recommending decisions for fire risk **management**. For example, the database of the fire risk …

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Artificial intelligence for **forest fire prediction**

GE Sakr, IH Elhajj, G Mitri… - 2010 IEEE/ASME …, 2010 - ieeexplore.ieee.org

… A **forest fire** risk zone **map** was constructed using a four-category risk scale. The … **spread** of **wild land fire** in a set of interacting automata. The system was able to simulate the fire **spread** …

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Social drivers of vulnerability to **wildfire** disasters: A **review** of the literature

N Lambrou, C Kolden, A Loukaitou-Sideris… - Landscape and Urban …, 2023 - Elsevier

… exposure and frequently overlooked the individual and community vulnerability to **wildfire**. Here, we **review** the emergent literature on social vulnerability to **wildfire** by synthesizing …

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A **review** of the main driving factors of **forest fire** ignition over Europe

A Ganteaume, A Camia, M Jappiot… - … **management**, 2013 - Springer

… , with increasing mobility of people, **spread** of tourism and recreational activities and the … , fire is a traditional and long established **management** tool. Its ancient and current use in …

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Modelling the spatial variability of **wildfire susceptibility** in Honduras using remote sensing and geographical information systems

MC Valdez, KT Chang, CF Chen… - … , Natural Hazards and …, 2017 - Taylor & Francis

… **Wildfire dynamics** and a countrywide **wildfire susceptibility map** are difficult to obtain, hence its spatial variability is generally inaccurate or it is unknown in many specific areas. For this …

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